

This is Article Title

Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to use this template in its entirety including structural elements, section heading, subheadings, citations, or equations.

At the end of the template, you need to provide information to the sections entitled Author Contributions, Data Availability, Code Availability, Declarations, Conflict of Interest, Financial Interests and Computing Interests.

You should read and save this LaTeX file. It contains almost all the elements you need to learn in order to write in LaTeX. Some free web services are also available. See ChatGPT, Overleaf.com and <https://www.mathcha.io/>

Keywords: keyword1, Keyword2, Keyword3, Keyword4

1 Introduction

The Introduction section, of referenced text [1] expands on the background of the work (some overlap with the Abstract is acceptable). The introduction should not include subheadings.

Springer Nature does not impose a strict layout as standard however authors are advised to check the individual requirements for the journal they are planning to submit to as there may be journal-level preferences. When preparing your text please also be aware that some stylistic choices are not supported in full text XML (publication version), including coloured font. These will not be replicated in the typeset article if it is accepted.

The paper is outlined as follows. Section 2 presents the problem statement. Section 3 discusses the method of solution. Section 4 shows numerical results in the table format. Section 5 shows figures. Section 6 presents Algorithms, Program Codes and Listings. Finally, Section 11 concludes the paper.

Required Citation Style

- You must cite equations this way: Eq. (1) or Eqs. (1) and (2).
- You must cite figures this way: Figure 1 or Figures 1 and 2.
- You must cite tables this way: Table 1 or Tables 1 and 2.
- You must cite references this way: New method is reported in [1]. New method and results are reported in [1,2].

2 Problem Statement Section

The main section of problem statement.

2.1 An example for second level head

This is a subsection of the problem statement. **We don't recommend to use deeper level of sub-sections such as Subsubsection 2.1.1.**

3 Method of Solution

Equations in L^AT_EX can either be inline or on-a-line by itself (“display equations”). For inline equations use the `$...$` commands. E.g.: The equation $H\psi = E\psi$ is written via the command `$H \psi = E \psi$`.

For display equations (with auto generated equation numbers) one can use the `equation` or `align` environments:

$$\|\tilde{X}(k)\|^2 \leq \frac{\sum_{i=1}^p \|\tilde{Y}_i(k)\|^2 + \sum_{j=1}^q \|\tilde{Z}_j(k)\|^2}{p+q}. \quad (1)$$

where,

$$\begin{aligned} D_\mu &= \partial_\mu - ig \frac{\lambda^a}{2} A_\mu^a \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c \end{aligned} \quad (2)$$

Notice the use of `\nonumber` in the `align` environment at the end of each line, except the last, so as not to produce equation numbers on lines where no equation numbers are required. The `\label{}` command should only be used at the last line of an `align` environment where `\nonumber` is not used.

$$Y_\infty = \left(\frac{m}{\text{GeV}} \right)^{-3} \left[1 + \frac{3 \ln(m/\text{GeV})}{15} + \frac{\ln(c_2/5)}{15} \right] \quad (3)$$

The class file also supports the use of `\mathbb{}`, `\mathscr{}` and `\mathcal{}` commands. As such `\mathbb{R}`, `\mathscr{R}` and `\mathcal{R}` produces $\mathbb{R}$, $\mathscr{R}$ and $\mathcal{R}$ respectively (refer Subsection 2.1).

4 Tables

Tables can be inserted via the normal `table` and `tabular` environment. To put footnotes inside tables you should use `\footnotetext[]{\dots}` tag. The footnote appears just below the table itself (refer Tables 1 and 2). For the corresponding footnote mark use `\footnotemark[\dots]`

In case of double column layout, tables which do not fit in single column width should be set to full text width. For this, you need to use `\begin{table*}` ...

Table 1 Caption text with a period.

Column 1	Column 2	Column 3	Column 4
row 1	data 1	data 2	data 3
row 2	data 4	data 5 ¹	data 6
row 3	data 7	data 8	data 9 ²

`\end{table*}` instead of `\begin{table} ... \end{table}` environment. Lengthy tables which do not fit in textwidth should be set as rotated table. For this, you need to use `\begin{sidewaystable} ... \end{sidewaystable}` instead of `\begin{table*} ... \end{table*}` environment. This environment puts tables rotated to single column width. For tables rotated to double column width, use `\begin{sidewaystable*} ... \end{sidewaystable*}`.

5 Figures

As per the $\text{\LaTeX}$ standards you need to use eps images for $\text{\LaTeX}$ compilation and pdf/jpg/png images for PDF $\text{\LaTeX}$ compilation. This is one of the major difference between $\text{\LaTeX}$ and PDF $\text{\LaTeX}$. Each image should be from a single input .eps/vector image file. Avoid using subfigures. The command for inserting images for $\text{\LaTeX}$ and PDF $\text{\LaTeX}$ can be generalized. The package used to insert images in $\text{\LaTeX}$ /PDF $\text{\LaTeX}$ is the graphicx package. Figures can be inserted via the normal figure environment as shown in the below example:



Fig. 1 This is a wide empty box. **For the best visual effect, create figures in eps format with large font size such as 14pt.** This is an example of long caption this is an example of long caption this is an example of long caption.

In case of double column layout, the above format puts figure captions/images to single column width. To get spanned images, we need to provide `\begin{figure*} ... \end{figure*}`.

For sample purpose, we have included the width of images in the optional argument of `\includegraphics` tag. Please ignore this.

6 Algorithms, Program Codes and Listings

Packages `algorithm`, `algorithmicx` and `algpseudocode` are used for setting algorithms in $\text{\LaTeX}$ using the format:

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Table 2 Tables which are too long to fit, should be written using the “sidewaystable” environment as shown here.

Projectile	Element 1 ¹			Element ²		
	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 $\pm$ 12	780 A	1166	1239 $\pm$ 100
Element 4	500 A	961	922 $\pm$ 10	900 A	1268	1092 $\pm$ 40
Element 5	990 A	1168	1547 $\pm$ 12	780 A	1166	1239 $\pm$ 100
Element 6	500 A	961	922 $\pm$ 10	900 A	1268	1092 $\pm$ 40

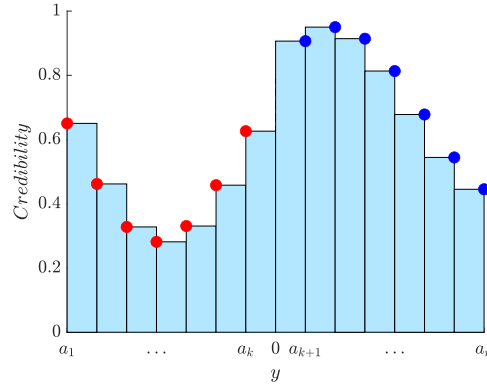


Fig. 2 End points used in computation of the expected value of a discrete fuzzy variable.

```

\begin{algorithm}
\caption{<alg-caption>}\label{<alg-label>}
\begin{algorithmic}[1]
. . .
\end{algorithmic}
\end{algorithm}

```

You may refer above listed package documentations for more details before setting `algorithm` environment. For program codes, the “verbatim” package is required and the command to be used is `\begin{verbatim} ... \end{verbatim}`.

Similarly, for `listings`, use the `listings` package. `\begin{lstlisting} ... \end{lstlisting}` is used to set environments similar to `verbatim` environment. Refer to the `lstlisting` package documentation for more details.

A fast exponentiation procedure:

```

begin
  for i:=1 to 10 step 1 do
    expt(2,i);
    newline() od
where
proc expt(x,n) ≡
  z:=1;
  do if n=0 then exit fi;
  do if odd(n) then exit fi;
    comment: This is a comment statement;
    n:=n/2; x:=x*x od;
  { n>0 };
  n:=n-1; z:=z*x od;
  print(z).
end

```

231 **Algorithm 1** Calculate $y = x^n$.

232 **Require:** $n \geq 0 \vee x \neq 0$

233 **Ensure:** $y = x^n$

234 1: $y \leftarrow 1$

235 2: **if** $n < 0$ **then**

236 3: $X \leftarrow 1/x$

237 4: $N \leftarrow -n$

238 5: **else**

239 6: $X \leftarrow x$

240 7: $N \leftarrow n$

241 8: **end if**

242 9: **while** $N \neq 0$ **do**

243 10: **if** N is even **then**

244 11: $X \leftarrow X \times X$

245 12: $N \leftarrow N/2$

246 13: **else**[N is odd]

247 14: $y \leftarrow y \times X$

248 15: $N \leftarrow N - 1$

249 16: **end if**

250 17: **end while**

```
for i:=maxint to 0 do
begin
{ do nothing }
end;
Write('Case_insensitive_');
Write('Pascal_keywords.');
```

262 7 Cross referencing

264 Environments such as figure, table, equation and align can have a label declared via
265 the `\label{#label}` command. For figures and table environments use the `\label{}`
266 command inside or just below the `\caption{}` command. You can then use the
267 `\ref{#label}` command to cross-reference them. As an example, consider the label
268 declared for Figure 1 which is `\label{FigWidefig}`. To cross-reference it, use the
269 command `Figure \ref{FigWidefig}`, for which it comes up as “Figure 1”.

270 To reference line numbers in an algorithm, consider the label declared for the line
271 number 2 of Algorithm 1 is `\label{algln2}`. To cross-reference it, use the command
272 `\ref{algln2}` for which it comes up as line 2 of Algorithm 1.

7.1 Details on reference citations

Standard L^AT_EX permits only numerical citations. To support both numerical and author-year citations this template uses natbib L^AT_EX package. For style guidance please refer to the template user manual.

Here is an example for \cite{...}: [1]. Another example for \cite{...}: [2]. All cited bib entries are printed at the end of this article: [3–7] and [8–13].

8 Examples for theorem like environments

For theorem like environments, we require amsthm package. There are three types of predefined theorem styles exists—thmstyleone, thmstyletwo and thmstylethree

thmstyleone	Numbered, theorem head in bold font and theorem text in italic style
thmstyletwo	Numbered, theorem head in roman font and theorem text in italic style
thmstylethree	Numbered, theorem head in bold font and theorem text in roman style

For mathematics journals, theorem styles can be included as shown in the following examples:

Theorem 1 (Theorem subhead). *Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Proposition 2. *Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Example 1. *Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Remark 1. *Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

323 **Definition 1** (Definition sub head). *Example definition text. Example definition text.*
 324 *Example definition text. Example definition text. Example definition text. Example*
 325 *definition text. Example definition text. Example definition text.*

326 Additionally a predefined “proof” environment is available: `\begin{proof} ...`
 327 `\end{proof}`. This prints a “Proof” head in italic font style and the “body text” in
 328 roman font style with an open square at the end of each proof environment.

329 *Proof.* Example for proof text. Example for proof text. Example for proof text. Exam-
 330 ple for proof text. Example for proof text. Example for proof text. Example for proof
 331 text. Example for proof text. Example for proof text. Example for proof text. □
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333 Sample body text. Sample body text. Sample body text. Sample body text. Sample
 334 body text. Sample body text. Sample body text. Sample body text.

335 *Proof of Theorem 1.* Example for proof text. Example for proof text. Example for
 336 proof text. Example for proof text. Example for proof text. Example for proof text.
 337 Example for proof text. Example for proof text. Example for proof text. Example for
 338 proof text. □
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340 For a quote environment, use `\begin{quote}...\end{quote}`

341 Quoted text example. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus
 342 volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque.
 343 Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis
 344 commodo.

345 Sample body text. Sample body text. Sample body text. Sample body text. Sample
 346 body text (refer Figure 1). Sample body text. Sample body text. Sample body text
 347 (refer Table 2).
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349 9 More about Methods

351 Topical subheadings are allowed. Authors must ensure that their Methods section
 352 includes adequate experimental and characterization data necessary for others in
 353 the field to reproduce their work. Authors are encouraged to include RIIDs where
 354 appropriate.
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356 **Ethical approval declarations** (only required where applicable) Any article
 357 reporting experiment/s carried out on (i) live vertebrate (or higher invertebrates),
 358 (ii) humans or (iii) human samples must include an unambiguous statement within
 359 the methods section that meets the following requirements:

- 360 1. Approval: a statement which confirms that all experimental protocols were
 361 approved by a named institutional and/or licensing committee. Please identify the
 362 approving body in the methods section
- 363 2. Accordance: a statement explicitly saying that the methods were carried out in
 364 accordance with the relevant guidelines and regulations
- 365 3. Informed consent (for experiments involving humans or human tissue samples):
 366 include a statement confirming that informed consent was obtained from all
 367 participants and/or their legal guardian/s
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10 Discussions	369
Discussions should be focused to reveal deeper understanding of the research results.	370
This part is not mandatory.	371
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11 Conclusion	374
Conclusions may be used to restate your hypothesis or research question, restate your	375
major findings, explain the relevance and the added value of your work, highlight any	376
limitations of your study, describe future directions for research and recommendations.	377
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References	380
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415 Appendix A Section title of first appendix

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417 An appendix contains supplementary information that is not an essential part of the
418 text itself but which may be helpful in providing a more comprehensive understanding
419 of the research problem or it is information that is too cumbersome to be included in
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430
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